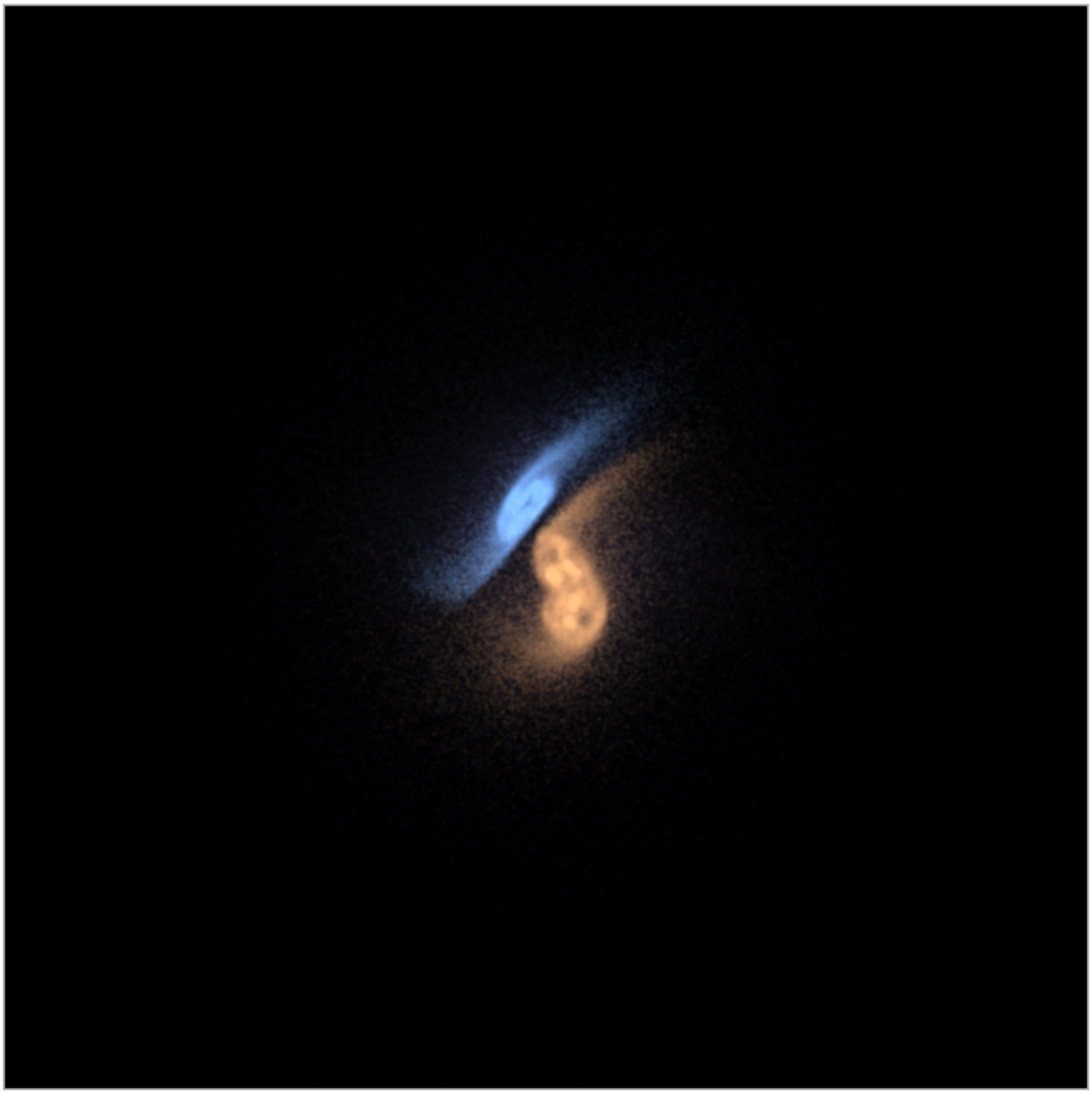
When Two Galaxies Collide

A direct N-body simulation on the Aurora exascale supercomputer



Just after first pericentre, t = 0.14 Gyr

This report follows 600,000 gravitating particles as two disk galaxies fall together, swing past each other, and begin to merge. Nothing in the simulation knows what a spiral arm or a tidal tail is. The only rule is Newton's law of gravitation, applied between every pair of particles at every step. The structures you see emerge from that rule alone.

Argonne Leadership Computing Facility | Aurora
Allocation: DD_plasma_physics

1. What the computer is actually doing

One rule, applied a great many times

Every particle feels a pull from every other particle. For particle i, the acceleration is the sum of the pulls from all the others:

$$\mathbf{a}_i = G\sum_j m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}$$

That is the whole physical model. There is no fluid, no magnetic field, no chemistry - just gravity. The difficulty is purely one of counting: with N = 600,000 particles, each step requires $N^2 \approx 0.36$ trillion pair interactions, and the simulation takes tens of thousands of steps.

Why the particles are not stars

A real galaxy holds something like 10^{11} stars, far more than we can track. Each particle here stands in for roughly a million stars' worth of mass. This is fine for studying the shape of a galaxy, because the large-scale gravitational field depends on where the mass is, not on how finely it is chopped up.

Softening: blurring the point masses

If two particles pass very close, the $1/r^2$ force diverges and the simulation blows up. Since our particles are stand-ins for large clouds of stars rather than true points, we blur each one slightly, replacing r^2 with $r^2 + \varepsilon^2$:

$$\mathbf{a}_i = G\sum_j m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{\left(|\mathbf{r}_j - \mathbf{r}_i|^2 + \varepsilon^2\right)^{3/2}}$$

Here $\varepsilon = 0.08$ kpc, about 260 light years. Below that scale the simulation deliberately does not resolve structure. A pleasant side effect: when j = i the numerator is zero, so a particle exerts no force on itself and no special case is needed in the code.

Stepping forward in time

We use the leapfrog (kick-drift-kick) scheme: nudge the velocities by half a step, move the positions a full step, recompute the forces, then nudge the velocities again. Leapfrog is symplectic - it respects the geometric structure of mechanics - so the total energy wanders slightly but never drifts steadily away, even over tens of thousands of steps. A naive Euler scheme would slowly inflate the galaxies into nonsense.

2. How the two galaxies were built

Each galaxy: a halo and a disk

Every galaxy is assembled from two pieces. A Plummer sphere of 4×10^{10} solar masses with a 20 kpc scale radius plays the part of the dark matter halo and central bulge; it is sampled exactly from its own distribution function, so it would sit still forever if left alone. On top sits an exponential stellar disk of 5×10^{10} solar masses with a 3.5 kpc scale length and 0.3 kpc thickness, with stars placed on near-circular orbits set by the mass enclosed inside them.

The disk is the part you can see. The halo is invisible in real observations and nearly invisible in the images here, but it carries most of the mass and therefore controls the encounter.

The orbit

The galaxies start 60 kpc apart on a parabolic orbit - exactly the marginally-bound case, the textbook idealisation of two galaxies falling together from rest at infinity. Their transverse velocity is set so that two point masses of the same total mass would pass within 12 kpc. Real galaxies are not point masses, though - their mass is spread over tens of kiloparsecs - so the true closest approach comes out larger than that estimate. Section 4 shows what actually happened. One disk is tilted 20° to the orbital plane and the other 70°, so the encounter is close to prograde for the first galaxy and strongly inclined for the second. That asymmetry is deliberate: prograde encounters, where the stars orbit in the same sense as the companion's approach, produce by far the longest tidal tails.

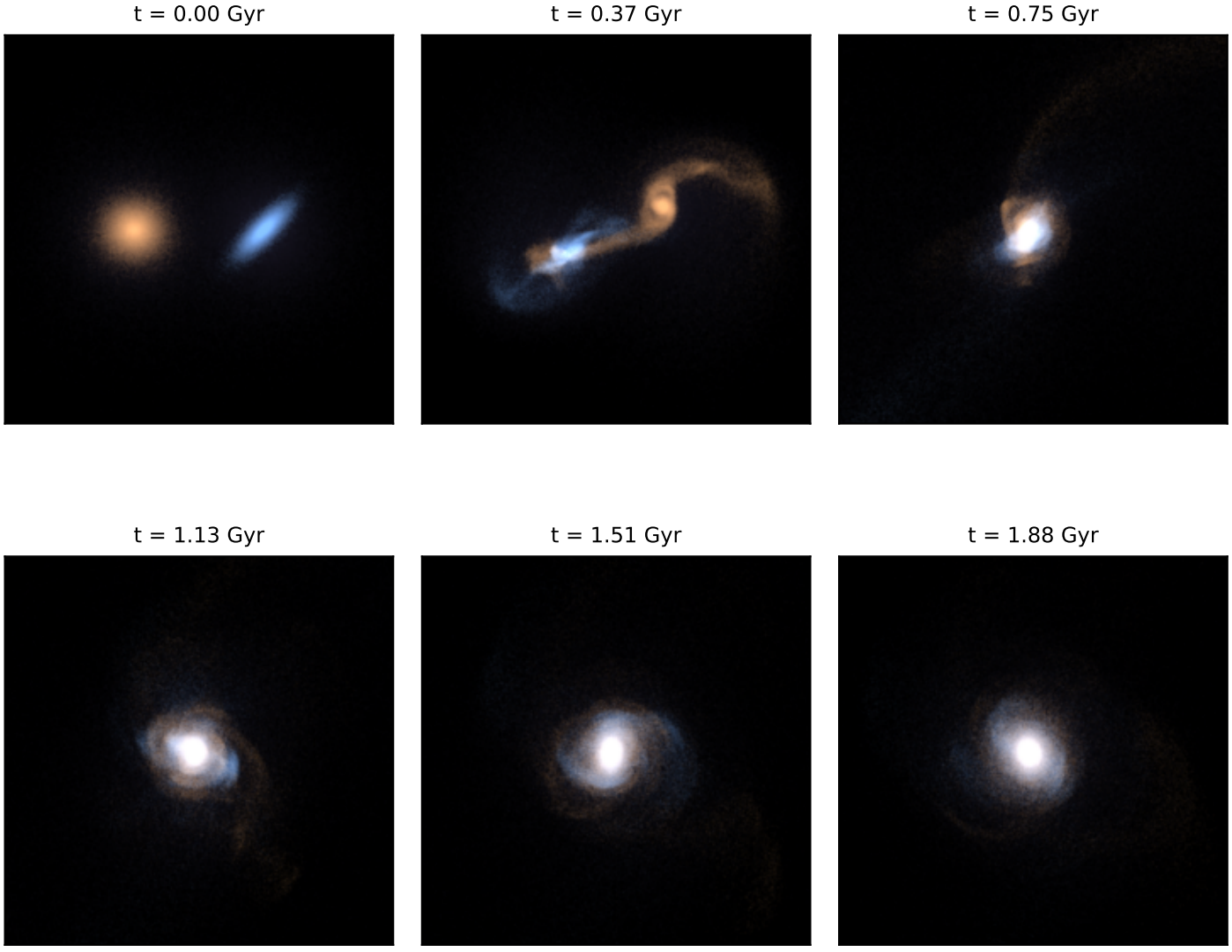
Units

The code works in units where G = 1, which keeps the numbers near one and avoids wasting floating-point precision. Converting back:

```
length    1 unit  =  1 kpc           =  3.26 thousand light years
mass      1 unit  =  10^10 M_sun     =  10 billion suns
velocity  1 unit  =  207.4 km/s
time      1 unit  =  4.715 Myr        =  4.7 million years
```

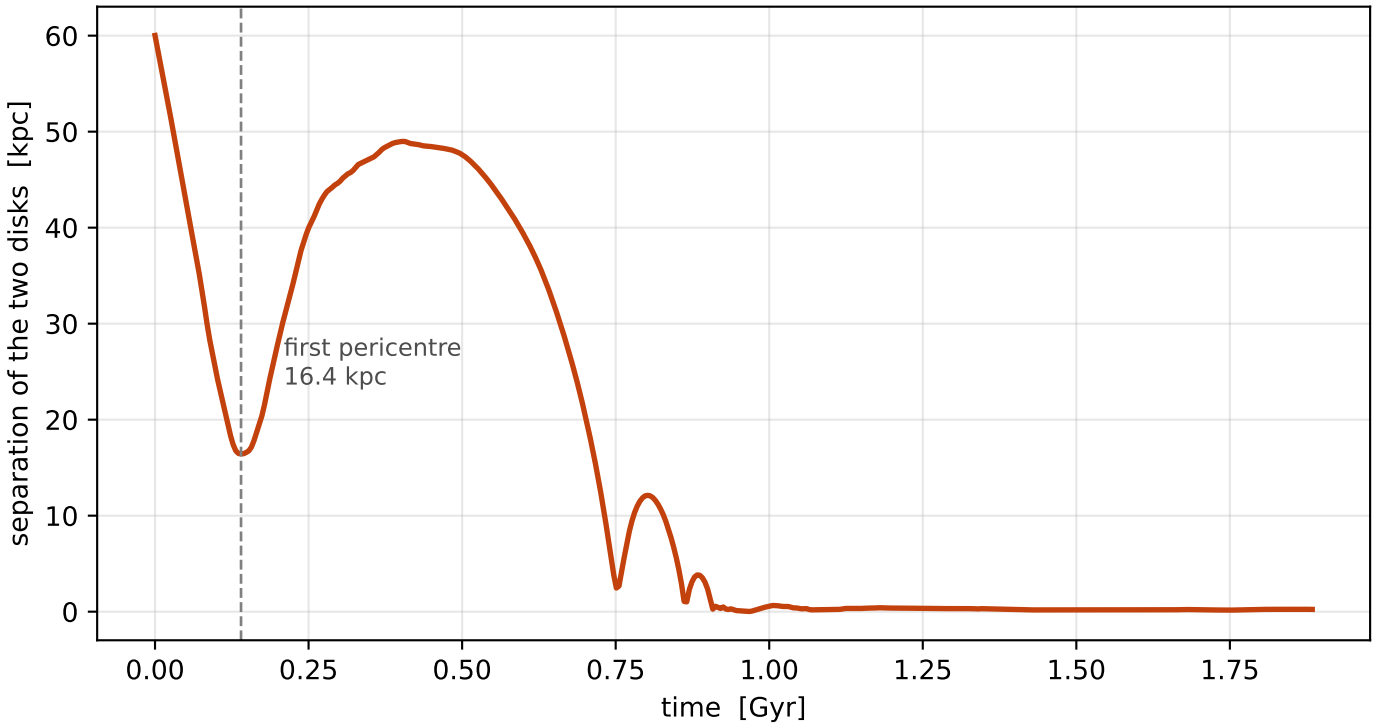
For scale, the Sun orbits the Milky Way once every ~230 million years, so the whole simulation below covers only a handful of galactic years.

3. The encounter, start to finish



Orange marks stars that began in the first galaxy, blue the second; the faint violet haze is dark matter. Watch for the bridge of material that forms between the two disks at closest approach, and the long tails flung outward on the opposite sides. Both are made of stars that were merely orbiting quietly beforehand - the companion's gravity pulls harder on the near side of a disk than on the far side, and that difference across the galaxy, the tidal force, is what tears the material out.

4. Reading the merger from one number



The distance between the centres of the two stellar disks tells most of the story. The galaxies fall together, reach 16 kpc at first pericentre around t = 0.14 Gyr, and swing apart again. Each close passage converts orbital energy into the random motions of individual stars - energy drained out of the orbit and into the tidal tails and the thickening of the disks. Because that drain is one-way, the galaxies cannot climb back to their original separation, so each apocentre is lower than the last and the pair spirals in. This is dynamical friction, and it is why galaxy encounters almost always end in a merger rather than a clean flyby. The end state, a single pressure-supported blob with no ordered rotation, is how astronomers think many elliptical galaxies were made.

A caution about what this model leaves out

There is no gas here, and therefore no star formation, no shocks and no feedback. Real colliding galaxies light up with newborn stars exactly where this simulation shows the densest tidal features, which is why the Antennae galaxies glow blue along their bridge. Treat these images as a map of where the mass goes, not a picture of what a telescope would see.

5. How it was computed

The machine

Aurora is an exascale supercomputer at Argonne National Laboratory. Each node carries six Intel Data Center GPU Max 1550 accelerators, and each of those is built from two tiles. With the frameworks module loaded the runtime presents every tile as its own device, so a node offers twelve. This run launched six MPI ranks, one per tile, and so used six of the twelve available tiles - twelve ranks would have finished in roughly half the time. Each rank holds a full copy of all particle positions and computes the forces on its own contiguous slice. After every step the slices are stitched back together with an MPI_Allgatherv, so the forces are always evaluated against the complete system.

Why direct summation

Most production N-body codes use a tree or a particle-mesh method to avoid the N^2 cost, approximating distant groups of particles by their combined mass. This code does not: it computes every pair exactly. That is affordable on GPUs at this particle count, and it means the only sources of error are the softening length and the timestep, both of which are easy to reason about. The inner loop stages particles through GPU local memory in tiles of 256 so that each one is fetched once and reused across the whole tile.

Measured performance

```
done: 20000 steps in 1977.4 s  (98.87 ms/step, 3.64e+12 interactions/s)
```

Reproducing this

```
cd /lus/flare/projects/DD_plasma_physics/claude-galaxies
make                                    # build with mpicxx -fsycl
./venv/bin/python make_ic.py --n 600000 --out ic.bin
qsub run_all.sh            # simulate, render, encode, report
```

Files

```
nbody.cpp      SYCL + MPI direct N-body solver
make_ic.py     builds the two galaxies and their orbit
render.py      snapshots -> PNG frames
report.py      this document
run_all.sh     the PBS job that produced everything here
config.sh      run parameters shared by the stages
collision.mp4  the animation
```